

the **acrosomal reaction** in the sperm. This reaction begins with the discharge of hydrolytic enzymes from the **acrosome**, a specialized vesicle at the tip of the sperm. These enzymes partially digest the jelly coat, enabling a sperm structure called the *acrosomal process* to form, elongate, and penetrate the coat. Protein molecules on the tip of the acrosomal process bind to specific proteins in the egg plasma membrane. This “lock-and-key” recognition is especially important for sea urchins and other species with external fertilization because the water into which sperm and eggs are released may contain gametes of other species (see, for example, Figure 24.3h).

The recognition event between the sperm and egg triggers fusion of their plasma membranes. The sperm nucleus enters the egg cytoplasm as ion channels open in the egg’s plasma membrane. Sodium ions diffuse into the egg and cause depolarization, a decrease in the charge difference, or potential, across the plasma membrane (see Concept 7.4).

Once depolarization takes place, additional sperm cannot fuse with the egg’s plasma membrane. In this way, depolarization acts as a barrier to **polyspermy**, the entry of multiple sperm nuclei into the egg. If polyspermy were to occur, the resulting abnormal number of chromosomes would be lethal for the embryo. Because depolarization occurs 1–3 seconds after a sperm binds to an egg, depolarization acts as a *fast block to polyspermy*.

The Cortical Reaction

Although membrane depolarization in sea urchins lasts for only a minute or so, there is a longer-lasting change that also prevents polyspermy. This *slow block to polyspermy* is established by vesicles in the outer rim, or cortex, of the cytoplasm. Within seconds after a sperm binds to the egg, these vesicles, called cortical granules, fuse with the egg plasma membrane (see Figure 47.2). Contents of the cortical granules are released into the space between the plasma membrane and the surrounding vitelline layer, a structure formed by the egg’s extracellular matrix. Enzymes and

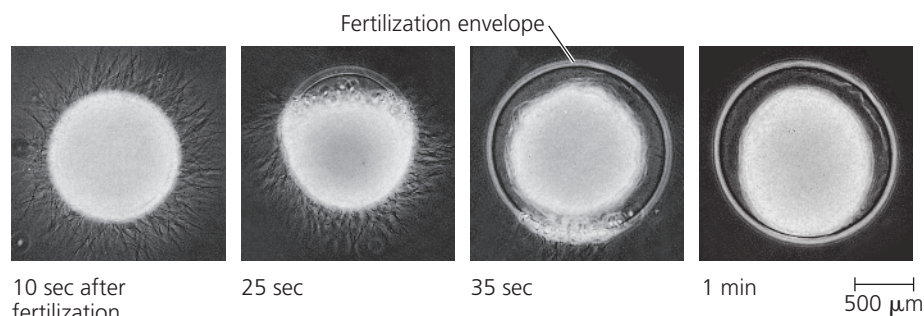
other granule contents then trigger a *cortical reaction*, which lifts the vitelline layer away from the egg and hardens the layer into a protective fertilization envelope.

Formation of the fertilization envelope requires a high concentration of calcium ions (Ca^{2+}) in the egg. As described in Figure 47.3, researchers wondered whether a change in the Ca^{2+} concentration triggers the cortical reaction. Using a

▼ Figure 47.3 Inquiry

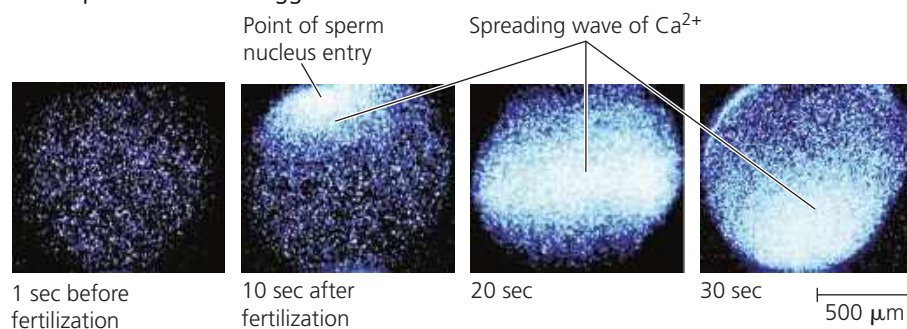
Does the distribution of Ca^{2+} in an egg correlate with formation of the fertilization envelope?

Experiment Investigators mixed sea urchin eggs with sperm, waited 10–60 seconds, and then added a chemical fixative, freezing cellular structures in place. When photographs of each sample are ordered according to the time of fixation, they illustrate the stages in the formation of a fertilization membrane for a single egg.



Calcium ion (Ca^{2+}) signaling controls fusion of vesicles with the plasma membrane during neurotransmitter release, insulin secretion, and plant pollen tube formation. Researchers hypothesized that Ca^{2+} signaling plays a similar role in forming the fertilization envelope. To test this hypothesis, they tracked the release of free Ca^{2+} in sea urchin eggs after sperm binding. A fluorescent dye that glows when it binds free Ca^{2+} was injected into unfertilized eggs. The scientists then added sea urchin sperm and used fluorescence to produce the results shown here.

Results A rise in the concentration of Ca^{2+} in the cytosol began near where the sperm entered and spread in a wave. Soon after the wave passed, the fertilization envelope rose from the egg surface.



Conclusion Ca^{2+} release correlates with formation of the fertilization envelope, supporting the researchers’ hypothesis that an increase in the Ca^{2+} level triggers cortical granule fusion.

Data from R. Steinhardt et al., Intracellular calcium release at fertilization in the sea urchin egg, *Developmental Biology* 58:185–197 (1977); M. Hafner et al., Wave of free calcium at fertilization in the sea urchin egg visualized with Fura-2, *Cell Motility and the Cytoskeleton* 9:271–277 (1988).

➔ **Instructors:** A related Experimental Inquiry Tutorial can be assigned in **Mastering Biology**.

WHAT IF? Suppose a particular molecule could enter the egg and bind to Ca^{2+} , blocking its function. How would you use this molecule to further test the hypothesis that a rise in the Ca^{2+} level triggers cortical granule fusion?